

Agentic Multimodal RAG with Grounded Explanations for Complex Scientific Reasoning

September 23, 2026

Abstract

As the volume of scientific literature grows, researchers increasingly rely on AI-assisted literature review systems. However, current multimodal Retrieval-Augmented Generation (RAG) models operate on a passive, single-pass paradigm, leading to severe factual hallucinations on complex, multi-hop scientific queries. Furthermore, these systems act as “black boxes,” providing no verifiable visual grounding for their claims. In this thesis, we propose an Agentic Multimodal RAG framework that utilizes an iterative reasoning loop to dynamically fetch text, tables, and figures. By integrating a strict Evidence Sufficiency Gate and a Grounded Explainability Module, our system generates precise spatial bounding boxes and retrieval execution logs. We outline a methodology to benchmark this framework on the SPIQA dataset, establishing a new standard for trust and verifiability in AI-assisted scientific document understanding.

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Chapter 1

Introduction

Researchers increasingly rely on AI-assisted literature review systems powered by Large Vision-Language Models (LVLMs) to synthesize information. However, scientific papers are highly multimodal. Vital findings are distributed across interleaved narrative text, dense statistical tables, and experimental plots. Comprehending these documents requires multi-hop reasoning across these distinct modalities. The motivation of this research stems from the urgent need to bridge the gap between text-only language models and the deeply multimodal reality of scientific research.

1.1 Problem Statement

Current multimodal Question Answering (QA) and Retrieval-Augmented Generation (RAG) systems fail on complex scientific literature due to two main reasons:

1. **Passive Retrieval:** Existing pipelines retrieve context in a single pass. They lack the iterative agency to perform multi-step cross-modal navigation (e.g., jumping from a text claim to a supporting table), leading to severe factual hallucinations.
2. **The Explainability Gap:** Current systems output answers with no verifiable visual grounding. They cannot point to the specific table cell or chart region used as evidence, making their generated reviews untrustworthy for academic use.

1.2 Research Questions

To address these limitations, this research is guided by the following questions:

- **RQ 1:** How does an iterative, agentic reasoning loop improve factual accuracy and reduce hallucination on the SPIQA dataset compared to single-pass retrieval?
- **RQ 2:** How effectively can visual document embeddings (e.g., ColPali) be utilized by a multimodal agent to fetch necessary tables and figures?
- **RQ 3:** Can the agent generate precise visual attribution (bounding boxes) for its answers, and how does this affect human verifiability?

1.3 Research Objectives

Mapping to the research questions, this work aims to achieve the following:

- **Objective 1:** Develop a Multimodal Agentic Planner that iteratively fetches text and figures.
- **Objective 2:** Integrate a vision-document retriever to dynamically fetch ground-truth visual data from PDFs.
- **Objective 3:** Implement an Evidence Sufficiency Gate to verify proof before answering, mitigating hallucinations.
- **Objective 4:** Engineer the output layer to generate spatial bounding-box coordinates on charts and tables to ensure verifiability.

1.4 Contributions

By bridging visual document retrieval with iterative agentic reasoning, this paper contributes a system that:

- **Outperforms Baselines:** Achieves superior QA accuracy on multi-hop scientific inquiries.
- **Eliminates Hallucinations:** Employs active sufficiency checks to ensure answers rely strictly on retrieved context.
- **Mitigates the Trust Gap:** Generates fine-grained visual attributions (exact bounding boxes) and an auditable retrieval log for every claim, establishing a new standard for academic verifiability.

Chapter 2

Methodology

This chapter describes how the proposed system works and how it will be evaluated. The presentation is deliberately ordered from intuition to formalism: each mechanism is first described in plain terms, then stated precisely. Every equation is followed by a reading of what it says in words, so that the formal notation documents the design rather than substituting for an explanation of it.

2.1 The Central Idea

A conventional Retrieval-Augmented Generation system behaves like a student who visits the library exactly once. Whatever material is gathered in that single trip is all the material available when the essay is written. If a claim turns out to need a supporting table that was not collected, the student cannot go back — and the most likely outcome is that the missing number is invented rather than the gap admitted.

The proposed system behaves instead like a careful researcher. It gathers some evidence, stops to ask whether that evidence is actually enough, returns for more when it is not, and refuses to write until the question can be answered from what it has in hand. Two mechanisms make this behaviour concrete:

1. An **agent** that decides, at each step, which kind of evidence to fetch next — and may fetch repeatedly rather than once.
2. An **Evidence Sufficiency Gate** that stands between gathering and writing, and only opens when the gathered evidence can support an answer.

The remainder of this chapter formalises these two mechanisms, describes the retrieval tools available to the agent, explains how evidence from different modalities is combined, and defines the evaluation plan.

2.2 Agentic System Architecture

2.2.1 The System as a State Machine

The agent is built on the ReAct paradigm, in which a model alternates between reasoning about what it needs and acting to obtain it. Formally, the system is a state machine:

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \mathcal{E} \rangle \quad (2.1)$$

In words, the system is fully described by five things: what it currently knows, what it is able to do, how doing something changes what it knows, what it gets back when it acts, and how it decides when to stop. Each component is defined below.

$\mathcal{S}$ — **the state**. Everything the system currently knows. This comprises the interpreted intent of the query, the record of which retrievals have already been performed, and the evidence buffer $\mathcal{B}_t$ holding every piece of evidence gathered so far.

$\mathcal{A}$ — **the action space**. The set of tools the agent may invoke. Each tool retrieves a different modality.

$\mathcal{T}$ — **the transition function**. How the state changes once a tool has been used: new evidence is added to the buffer and the retrieval history is extended.

$\mathcal{O}$ — **the observation space**. What comes back from a tool — a text passage, a parsed table, or a page region.

For example, the query “Does the accuracy gain reported in Section 4 hold for the low-resource subset?” decomposes into two sub-goals: first locate the claim about the accuracy gain in the narrative text, then locate the table that reports results broken down by subset. Decomposition matters because the second sub-goal cannot even be formulated until the first has been answered — it is the retrieved claim that tells the agent which table to look for. This is what is meant by *multi-hop* reasoning, and it is the reason a single retrieval pass is structurally incapable of answering such questions.

2.2.3 The Reasoning Loop

At each time step t the agent selects one action a_t from the action space:

$$\mathcal{A} = \{\text{TextRetriever}, \text{TableRetriever}, \text{FigureRetriever}\} \quad (2.3)$$

Executing a_t returns an observation o_t , which is appended to the evidence buffer. After t steps the buffer therefore contains the accumulated result of every retrieval performed so far:

$$\mathcal{B}_t = \bigcup_{i=1}^t o_i \quad (2.4)$$

The union notation simply says that nothing is discarded: evidence retrieved at step 1 is still available at step 5. The agent then consults the Sufficiency Gate, and either loops back to select another tool or proceeds to generation.

2.2.4 The Evidence Sufficiency Gate

The Sufficiency Gate is the principal novelty of the architecture, and its purpose is best stated as a question the system asks itself before writing anything: “*Can I answer this from what I have actually retrieved?*”

Concret

fluent, confident, unsupported answers. The gate removes the opportunity to make that error rather than attempting to detect it afterwards.

The loop is bounded by a maximum iteration count. If sufficiency is not reached before the bound is exhausted, the system reports that the evidence is inadequate rather than answering anyway. For a literature review tool this is the correct failure behaviour: a researcher can act on a stated gap, but cannot detect a fabrication.

2.3 Modality-Specific Retrieval Tools

Scientific PDFs are not uniform documents, and a single retrieval mechanism handles them poorly. The action space therefore provides three tools, each matched to a different kind of content. Each is described below together with the reason it exists as a separate tool.

TextRetriever. Performs dense semantic retrieval over chunked narrative text using scientific embeddings such as SciBERT. Domain-specific embeddings are used rather than general-purpose ones because scientific vocabulary is precisely where general embeddings are weakest.

TableRetriever. A structured parser that converts tables into a normalised form *while preserving the two-dimensional coordinates of every cell*. Coordinate preservation is not incidental detail. The bounding boxes required by Objective 4 can only be produced if the spatial layout of the table survives parsing; a parser that returned a table as flat text would satisfy the reasoning requirement and silently destroy the attribution requirement.

FigureRetriever (ColPali). Embeds entire PDF pages as grids of visual patches using vision-language multi-vector retrieval, locating charts, plots and diagrams without depending on OCR. This matters because OCR fails most severely exactly where scientific documents are most information-dense — multi-column headers, merged cells, and axis labels. Treating the page as an image sidesteps that failure mode entirely.

2.4 Cross-Modal Evidence Fusion

Once evidence has been retrieved from different modalities, the system must relate the pieces to one another. A symbol appearing in a sentence and the same symbol appearing in a chart legend must be recognised as referring to the same

quantity; otherwise the system holds two disconnected facts rather than one linked piece of evidence.

This linkage is performed by a cross-attention layer, in which the text representation queries the visual representation:

$$\mathbf{Z}_{\text{fused}} = \text{Softmax} \left(\frac{\mathbf{Q}_t \mathbf{K}_v^\top}{\sqrt{d_k}} \right) \mathbf{V}_v + \mathbf{H}_t \quad (2.5)$$

Read plainly, this says: *for each text token, measure how strongly it matches each visual patch; convert those match scores into weights; use the weights to build a summary of the visual evidence most relevant to that token; and add this summary to the token’s own representation.* The three factors play the following roles:

- $\mathbf{Q}_t \mathbf{K}_v^\top$ scores every text token against every visual patch — the raw compatibility between what the text says and what the page shows.
- Division by $\sqrt{d_k}$ keeps these scores in a numerically stable range before the softmax, which is standard practice in scaled dot-product attention.
- The final $+\mathbf{H}_t$ is a residual connection: the visual evidence *augments* the text representation rather than replacing it, so textual meaning is not lost when visual grounding is added.

The practical consequence is that downstream generation is conditioned on linked multimodal evidence rather than on two independent evidence streams that happen to have been retrieved together.

2.5 Grounded Multimodal Explainability

A system that produces a correct answer without showing where it came from remains unusable for academic work, because a reader has no way to check it. The generation phase therefore emits three audit artefacts alongside the natural-language response. Each answers a different verification question.

3. **Retrieval DAG Trace** — “*How did the system get here?*” A logged execution graph recording which tools were called, in what order, and how many times the Sufficiency Gate rejected the buffer before opening. This makes the reasoning path auditable rather than merely asserted.

Together these mean a reader need not trust the system’s answer. They can check it against the highlighted sentence, look at the boxed table cell, and inspect the path by which both were found.

2.6 Experimental Setup and Evaluation Plan

2.6.1 Dataset

The framework will be evaluated on **SPIQA**, a dataset of over 270,000 questions over scientific papers that require synthesis across figures, tables and text. SPIQA is appropriate here specifically because its questions are constructed to be unanswerable from any single modality, which is the condition under which the difference between iterative and single-pass retrieval becomes measurable at all.

2.6.2 Baselines

Two baselines isolate the two contributions of this work:

- **Single-Pass Multimodal RAG** (e.g. LLaVA-RAG) — multimodal but not iterative. Comparison against this baseline isolates the contribution of *agency*: what is gained by being permitted to retrieve more than once.
- **Text-Only Agentic Reasoning** (e.g. ReAct) — iterative but not multimodal. Comparison against this baseline isolates the contribution of *visual retrieval*: what is gained by being able to see tables and figures at all.

Because each baseline holds one contribution fixed and removes the other, an improvement over both cannot be attributed to either factor alone.

2.6.3 Evaluation Metrics

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Retrieval Faithfulness (RF). The precision of the retrieved evidence against the ground-truth evidence for each question. A low score indicates the system reached its answer without ever holding the material that justifies it. *Addresses RQ1 and RQ2.*

Modality Attribution Precision (MAP). How accurately generated content is traced back to the correct source modality. A low score indicates the system cannot reliably distinguish what it read in the text from what it read in a figure. *Addresses RQ2.*

Agent Efficiency Score (AES). The ratio of necessary retrieval hops to total tool calls. This guards against a degenerate solution in which the agent achieves high accuracy by retrieving indiscriminately; such a system would be accurate but neither efficient nor interpretable. *Addresses RQ1.*

Cross-modal Grounding Score (CGS). Intersection-over-Union between generated bounding boxes and ground-truth regions, measuring how precisely the system localises its visual evidence. *Addresses RQ3.*

Taken together, RF and AES measure whether the agentic loop works, MAP measures whether cross-modal retrieval works, and CGS measures whether the attribution is precise enough to be genuinely verifiable by a human reader.

Chapter 3

System Architecture

The proposed system is organized as a four-stage pipeline: an ingestion and indexing layer, an agentic planner, an evidence sufficiency gate, and an answer and attribution generation layer. Unlike passive single-pass retrieval pipelines, the planner and gate together form a closed feedback loop, allowing the system to iteratively gather additional evidence before committing to an answer.

3.1 Overview

Figure 3.1 illustrates the overall data flow. A user query enters the Agentic Planner, which issues retrieval actions against a dual index built during ingestion. After each retrieval round, the Evidence Sufficiency Gate evaluates whether the collected evidence is adequate to answer the query. If not, control returns to the planner for another retrieval iteration; if so, the system proceeds to generate a grounded answer with visual attribution.

3.2 Ingestion and Indexing Layer

Each PDF page in the corpus is rasterized into a page image, which is embedded using a visual document retriever (ColPali) to produce patch-level embeddings without requiring OCR or layout parsing. In parallel, narrative text is extracted and segmented into chunks, which are embedded using a dense text retriever. This produces two complementary indexes:

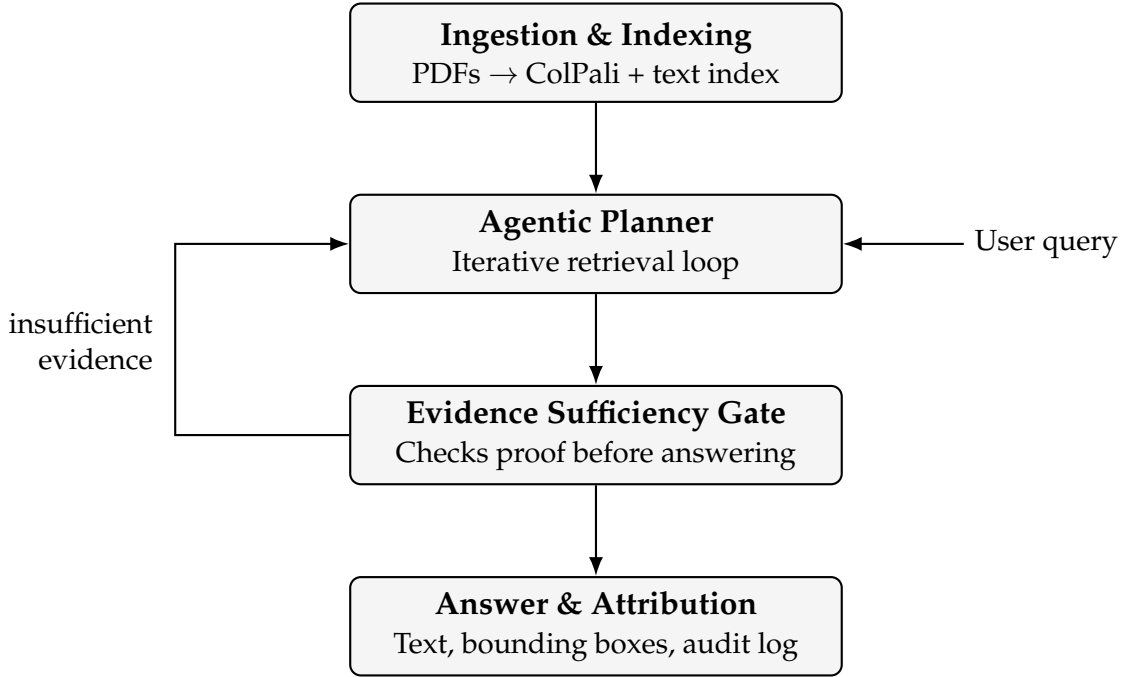


Figure 3.1: High-level architecture of the multimodal agentic literature review system.

- **Visual index:** Page-level patch embeddings for retrieving tables, figures, and dense visual content.
- **Textual index:** Chunk-level embeddings for retrieving narrative claims and discussion.

3.3 Agentic Planner

The Agentic Planner is an LVLM-driven controller that operates an iterative reasoning loop (Thought → Action → Observation), inspired by the ReAct paradigm. At each step, the planner selects one of the following tools:

- `retrieve_text(query)`: dense retrieval over the textual index.
- `retrieve_visual(query)`: visual retrieval over the ColPali index.
- `zoom_region(page_id, bbox)`: crops a sub-region of a page for closer inspection of dense tables or charts.
- `cross_reference(claim, entity)`: performs multi-hop navigation from a textual claim to a supporting table or figure.

3.4 Evidence Sufficiency Gate

After each retrieval round, the Evidence Sufficiency Gate determines whether the evidence accumulated in the scratchpad is sufficient to answer the query. Two candidate implementations are considered:

- **LLM-as-judge gate:** A prompted model evaluates the query and current evidence, returning a structured judgment of sufficiency and any missing information.
- **Learned gate:** A lightweight classifier trained on retrieval confidence scores and evidence coverage features.

If the gate reports insufficient evidence, control returns to the Agentic Planner for an additional retrieval iteration, bounded by a maximum iteration count to prevent unbounded looping. This mechanism directly implements Objective 3 and mitigates hallucination by preventing the system from answering without grounded proof.

3.5 Answer and Attribution Generation Layer

Once the gate confirms sufficiency, the system generates a final answer strictly conditioned on the retrieved evidence pool. Visual attribution is produced by leveraging the patch-level similarity scores from the ColPali retrieval step to localize the most relevant region of the source page, yielding bounding-box coordinates over the supporting table or figure. This satisfies Objective 4 and provides the fine-grained visual grounding required for human verifiability (RQ3).

3.6 Mapping to Research Objectives

Table 3.1 summarizes how each architectural component addresses the research objectives introduced in Chapter 1.

Component	Research Objective
Ingestion and Indexing Layer	Objective 2
Agentic Planner	Objective 1
Evidence Sufficiency Gate	Objective 3
Answer and Attribution Generation Layer	Objective 4

Table 3.1: Mapping of architectural components to research objectives.

Chapter 4

Graph Architecture and Node Specification

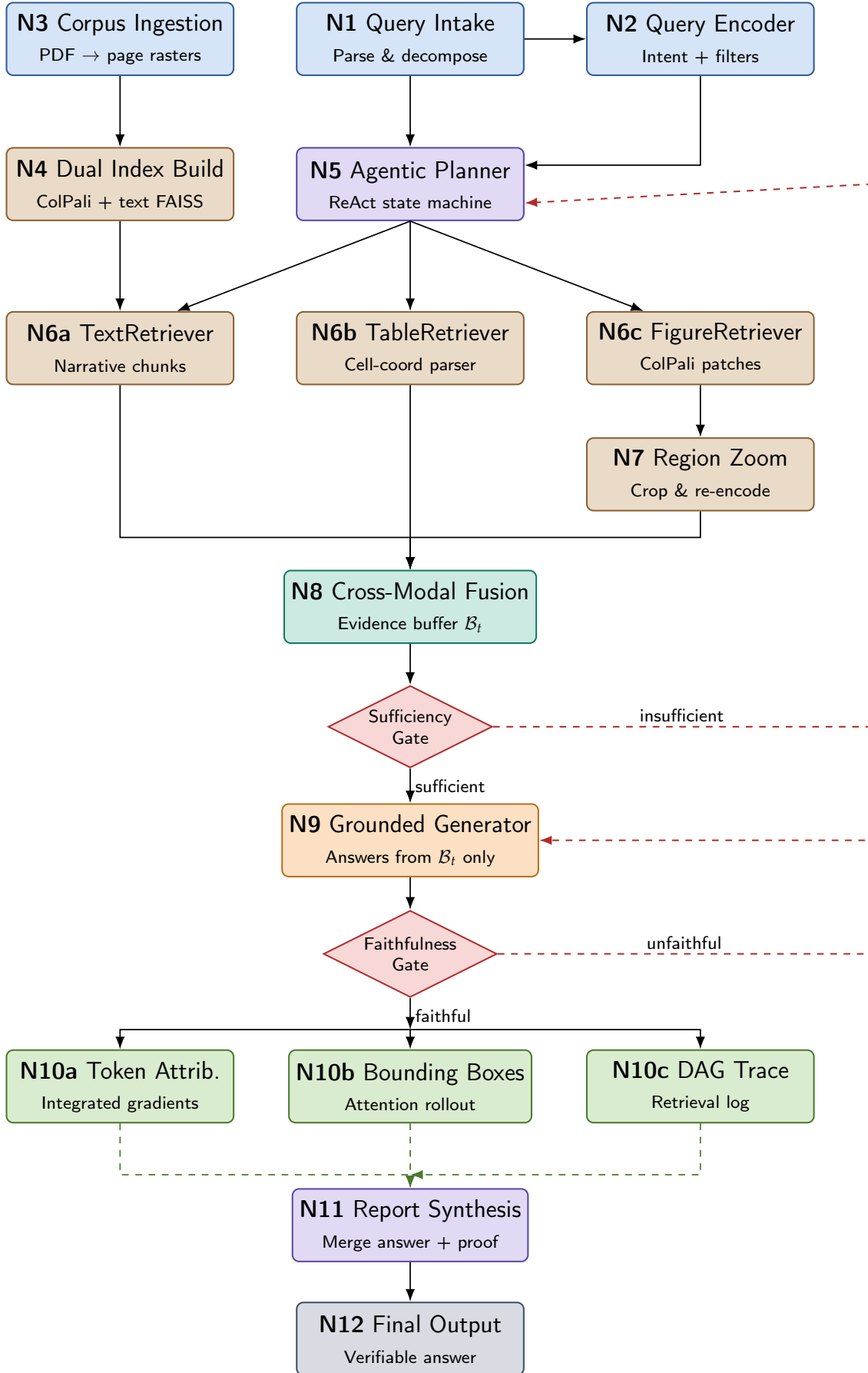
This chapter describes the system as a graph: a set of nodes, each with one job, connected by edges that show what flows between them. The goal is that this graph can be built node-for-node in a framework such as LangGraph, with nothing left to guess between what the thesis says and what the code does.

4.1 Design Rationale

Two design choices set this system apart from a standard single-pass RAG pipeline.

Retrieval can repeat. A standard pipeline searches once and answers once, so it only works if that one search happens to find everything needed. Here, the planner (N5) can call retrieval tools again and again, and the Sufficiency Gate sends it back until the evidence is actually enough. Multi-hop reasoning is not a prompting trick; it is simply how the loop is built.

Only certain nodes can add evidence. Nodes N6a–N7 are the only ones allowed to bring anything into the evidence buffer (N8), and the generator (N9) can only read from that buffer — it has no other way to see the corpus. This separation is what makes the Faithfulness Gate work: it only has to check claims against one fixed source of truth, instead of guessing whether the generator made something up.



4.2 Node-by-Node Specification

Here is how one query moves through the graph, start to finish.

Running example. *“Does the reported accuracy gain in Section 4 hold for the low-resource subset, and which table reports it?”*

This question needs two different kinds of evidence: a claim from the text and a number from a table. A single search cannot answer it alone, which is exactly why it is used here.

4.2.1 Ingestion Path (N3–N4)

This part runs once per corpus, offline, before any query arrives.

N3 — Corpus Ingestion. Every PDF page is converted into an image, and its text is separately pulled out and split into chunks. Pages become images rather than parsed layouts because tables are exactly where OCR fails — merged cells and multi-column headers break text extraction. An image keeps the table’s structure intact.

N4 — Dual Index Build. Two search indexes are built from the same set of pages: one holds visual embeddings of each page image (for finding tables and figures), the other holds text embeddings of each chunk (for finding claims). Because both indexes point back to the same pages, a match in one can lead directly to the matching spot in the other — which is what makes jumping from a text claim to its supporting table possible.

4.2.2 Query Encoding Path (N1–N2)

N1 — Query Intake. The question is broken into smaller steps before any search happens. For the running example, that is two steps: find the accuracy claim, then find the table that reports the low-resource breakdown. Doing this upfront gives the planner a plan to follow instead of improvising one.

N2 — Query Encoder. The query is embedded, and filters are pulled out of it — here, “Section 4” and “

4.2.3 Planning and Retrieval (N5–N7)

N5 — Agentic Planner. This is the only node that decides what happens next. It keeps track of the plan, what has already been searched, and what evidence has been found, then picks one retrieval tool to run. After seeing the result, it either searches again or moves on to the Sufficiency Gate. It never adds evidence itself — only a retrieval tool can do that.

N6a — TextRetriever. Searches the text index for narrative chunks matching the query. On the running example, this is the first search: it finds the sentence claiming the accuracy gain, along with the page it appears on.

N6b — TableRetriever. Extracts a table’s structure while keeping track of exactly where each cell sits on the page. This matters because the bounding boxes generated later (N10b) are only possible if this spatial position is kept — a plain-text version of the table would lose it.

N6c — FigureRetriever (ColPali). Searches the visual index to find charts, plots, or dense tables without needing OCR. It also keeps the match scores it computed, because those scores are reused later to draw bounding boxes.

N7 — Region Zoom. Crops in on part of a page and re-encodes it at higher resolution. This exists because finding the right page and reading a specific row inside it need different levels of detail — a page-level match tells you which page has the table, not which row answers the question. On the running example, this runs after N6c finds the right page, zooming into the low-resource row so its cells become readable.

4.2.4 Evidence Buffer and Sufficiency Gate (N8, G1)

N8 — Cross-Modal Fusion. Everything retrieved so far is converted into one shared format and added to the evidence buffer, along with where it came from — which page, which tool, which step. This buffer is the only thing the generator will ever see; nothing after this point looks at the corpus again.

Sufficiency Gate. Before generation

This check happens *before* generation, not after. A standard pipeline would hand the incomplete evidence to the generator and hope it notices something is missing — which is exactly how confident, wrong answers get produced.

4.2.5 Generation and Faithfulness Gate (N9, G2)

N9 — Grounded Generator. Writes the answer using only what is in the evidence buffer. It cannot search on its own and cannot fill in numbers from memory — everything it says has to trace back to something already retrieved.

Faithfulness Gate. Checks every factual claim in the draft answer against the evidence buffer. If the answer states something that is not actually backed by retrieved evidence, the gate fails and N9 is sent back to rewrite that part using only what is really there. Because the buffer is the only source allowed, this is a simple check against one fixed list of facts, not a judgement call.

4.2.6 Attribution and Output (N10–N12)

N10a — Token Attribution. Highlights which retrieved sentences most influenced each part of the answer.

N10b — Bounding Boxes. Draws a box — $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ — around the exact spot on the page a claim came from. This only works because N6b kept cell positions and N6c kept its match scores.

N10c — DAG Trace. Logs what actually happened: which tools ran, in what order, and how many times the Sufficiency Gate sent things back. On the running example, this log shows two search rounds and one rejection — so the process can be checked, not just trusted.

N11 — Report Synthesis. Combines the answer with its three proof artefacts into one readable report.

N12 — Final Output. What the user actually sees: the answer, the highlighted source text, a box on the source page

The retrieval loop (not enough evidence). Triggered by the Sufficiency Gate. Sends control back to N5 to search again. This stops the system from confidently answering using incomplete evidence.

The regenerate loop (wrong evidence used). Triggered by the Faithfulness Gate. Sends control back to N9 to rewrite. This stops the system from stating something the evidence does not actually support.

Both loops have a limit on how many times they can repeat. If that limit is reached, the system says the evidence is not enough instead of guessing. For a literature review tool, this is the right trade-off: “I could not confirm this” is useful. A confident wrong answer is not.

4.4 Mapping to Research Objectives

Table 4.1 shows which objective each part of the graph is responsible for.

Nodes	Objective	Contribution
N3, N4, N6c	Objective 2	Vision-document retrieval over rasterised pages
N1, N2, N5, N7	Objective 1	Iterative agentic planning and multi-hop navigation
N8, Sufficiency Gate	Objective 3	Evidence verification prior to generation
N9, Faithfulness Gate	Objective 3	Claim-level grounding of generated output
N10a–N10c, N11	Objective 4	Spatial, textual, and procedural attribution

Table 4.1: Mapping of graph nodes to research objectives.